

Lecture/Week 12

Numerical Methods in Linear AlgebraIterative Methods

- 1) Jacobi's Method
- 2) Gauss-Seidal Method.

Jacobi's Method

Given:- Linear System

$$\begin{array}{rcl}
 a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n & = & b_1 \\
 a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n & = & b_2 \\
 a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \dots + a_{3n}x_n & = & b_3 \\
 \vdots & & \vdots \\
 a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + a_{nn}x_n & = & b_n
 \end{array}$$

Step I:-

$$x_1 = \frac{1}{a_{11}} [b_1 - (a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n)]$$

$$x_2 = \frac{1}{a_{22}} [b_2 - (a_{21}x_1 + a_{23}x_3 + \dots + a_{2n}x_n)]$$

$$x_3 = \frac{1}{a_{33}} [b_3 - (a_{31}x_1 + a_{32}x_2 + \dots + a_{3n}x_n)]$$

$\vdots$

$$x_n = \frac{1}{a_{nn}} [b_n - (a_{n1}x_1 + a_{n2}x_2 + \dots + a_{n,n-1}x_{n-1})]$$

Condition

$$a_{ii} \neq 0$$

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Iterative Scheme Jacobi Method

Step II :-

$$x_1^{m+1} = \frac{1}{a_{11}} \left[b_1 - (a_{12}x_2^m + a_{13}x_3^m + \dots + a_{1n}x_n^m) \right]$$

$$x_2^{m+1} = \frac{1}{a_{22}} \left[b_2 - (a_{21}x_1^m + a_{23}x_3^m + \dots + a_{2n}x_n^m) \right]$$

$$x_3^{m+1} = \frac{1}{a_{33}} \left[b_3 - (a_{31}x_1^m + a_{32}x_2^m + a_{34}x_4^m + \dots + a_{3n}x_n^m) \right]$$

⋮

$$x_n^{m+1} = \frac{1}{a_{nn}} \left[b_n - (a_{n1}x_1^m + a_{n2}x_2^m + \dots + a_{nn-1}x_{n-1}^m) \right]$$

where ~~m=0,1,2,...~~ $m=0,1,2, \dots$

$$x^0 = [x_1^0, x_2^0, x_3^0, \dots, x_n^0] = [0, 0, 0, \dots, 0]$$

Iterative Scheme Gauss-Seidal Method

$$x_1^{m+1} = \frac{1}{a_{11}} \left[b_1 - (a_{12}x_2^m + a_{13}x_3^m + \dots + a_{1n}x_n^m) \right]$$

$$x_2^{m+1} = \frac{1}{a_{22}} \left[b_2 - (a_{21}x_1^{m+1} + a_{23}x_3^m + \dots + a_{2n}x_n^m) \right]$$

$$x_3^{m+1} = \frac{1}{a_{33}} \left[b_3 - (a_{31}x_1^{m+1} + a_{32}x_2^{m+1} + \dots + a_{3n}x_n^m) \right]$$

$$\left\{ \begin{array}{l} x_n^{m+1} = \frac{1}{a_{nn}} \left[b_n - (a_{n1}x_1^{m+1} + a_{n2}x_2^{m+1} + \dots + a_{nn-1}x_{n-1}^{m+1}) \right] \end{array} \right.$$

Example ①

$$\begin{aligned}5x_1 - 2x_2 + 3x_3 &= -1 \\ -3x_1 + 9x_2 + x_3 &= 2 \\ 2x_1 - x_2 - 7x_3 &= 3\end{aligned}$$

Use Jacobi Method to find the solution correct to 3-D places.

Solution

$$x_1 = \frac{1}{5} [-1 + 2x_2 - 3x_3]$$

$$x_2 = \frac{1}{9} [2 + 3x_1 - x_3]$$

$$x_3 = \frac{-1}{7} [3 - 2x_1 + x_2]$$

$$x_1^{m+1} = \frac{1}{5} [-1 + 2x_2^m - 3x_3^m]$$

$$x_2^{m+1} = \frac{1}{9} [2 + 3x_1^m - x_3^m]$$

$$x_3^{m+1} = \frac{-1}{7} [3 - 2x_1^m + x_2^m]$$

$$\begin{aligned}X^0 &= [x_1^0, x_2^0, x_3^0] \\ &= [0, 0, 0]\end{aligned}$$

For $m=0$

$$x_1^1 = \frac{1}{5} [-1 + 2x_2^0 - 3x_3^0] = \frac{1}{5} [-1 + 2(0) - 3(0)] = -\frac{1}{5} = -0.200$$

$$x_2^1 = \frac{1}{9} [2 + 3x_1^0 - x_3^0] = \frac{1}{9} [2 + 3(0) - (0)] = \frac{2}{9} = 0.222$$

$$x_3^1 = \frac{-1}{7} [3 - 2x_1^0 + x_2^0] = \frac{-1}{7} [3 - 2(0) + (0)] = -\frac{3}{7} = -0.429$$

For $m=1$

$$x_1^2 = \frac{1}{5} [-1 + 2x_2^1 - 3x_3^1] = \frac{1}{5} [-1 + 2(0.222) - 3(-0.429)] = 0.146$$

$$x_2^2 = \frac{1}{9} [2 + 3x_1^1 - x_3^1] = \frac{1}{9} [2 + 3(-0.200) - (-0.429)] = 0.203$$

$$x_3^2 = \frac{-1}{7} [3 - 2x_1^1 + x_2^1] = \frac{-1}{7} [3 - 2(0.200) + (-0.429)] = -0.517$$

For m=2

$$x_1^3 = \frac{1}{5} [1 + 2x_2^2 - 3x_3^2] = \frac{1}{5} [1 + 2(0.203) - 3(-0.517)] = 0.192$$

$$x_2^3 = \frac{1}{9} [2 + 3x_1^2 - x_3^2] = \frac{1}{9} [2 + 3(0.146) - (-0.517)] = 0.328$$

$$x_3^3 = \frac{-1}{7} [3 - 2x_1^2 + x_2^2] = \frac{-1}{7} [3 - 2(0.146) + (0.203)] = -0.416$$

For m=3

$$x_1^4 = \frac{1}{5} [1 + 2x_2^3 - 3x_3^3] = \frac{1}{5} [1 + 2(0.328) - 3(-0.416)] = 0.181$$

$$x_2^4 = \frac{1}{9} [2 + 3x_1^3 - x_3^3] = \frac{1}{9} [2 + 3(0.192) - (-0.416)] = 0.332$$

$$x_3^4 = \frac{-1}{7} [3 - 2x_1^3 + x_2^3] = \frac{-1}{7} [3 - 2(0.192) + (0.328)] = -0.421$$

For m=4

$$x_1^5 = 0.185$$

$$x_2^5 = 0.329$$

$$x_3^5 = -0.424$$

For m=5

$$x_1^6 = 0.186$$

$$x_2^6 = 0.331$$

$$x_3^6 = -0.423$$

For m=6

$$x_1^7 = 0.186$$

$$x_2^7 = 0.331$$

$$x_3^7 = -0.423$$

Ans.

Example (2)

$$5x_1 - 2x_2 + 3x_3 = -1$$

$$-3x_1 + 9x_2 + x_3 = 2$$

$$2x_1 - x_2 - 7x_3 = 3$$

Use Gauss-Seidal Method (5)

to find solution correct to 3-D places.

Solution

$$x_1 = \frac{1}{5} [-1 + 2x_2 - 3x_3]$$

$$x_2 = \frac{1}{9} [2 + 3x_1 - x_3]$$

$$x_3 = \frac{-1}{7} [3 - 2x_1 + x_2]$$

$$x_1^{m+1} = \frac{1}{5} [-1 + 2x_2^m - 3x_3^m]$$

$$x_2^{m+1} = \frac{1}{9} [2 + 3x_1^{m+1} - x_3^m]$$

$$x_3^{m+1} = \frac{-1}{7} [3 - 2x_1^{m+1} + x_2^{m+1}]$$

$$X_0 = [x_1^0, x_2^0, x_3^0] \\ = [0, 0, 0]$$

For $m=0$

$$x_1' = \frac{1}{5} [-1 + 2x_2^0 - 3x_3^0] = \frac{1}{5} [-1 + 2(0) - 3(0)] = -0.200$$

$$x_2' = \frac{1}{9} [2 + 3x_1' - x_3^0] = \frac{1}{9} [2 + 3(-0.200) - (0)] = 0.156$$

$$x_3' = \frac{-1}{7} [3 - 2x_1' + x_2'] = \frac{-1}{7} [3 - 2(-0.200) + (0.156)] = -0.508$$

For $m=1$

$$x_1^2 = \frac{1}{5} [-1 + 2x_2^1 - 3x_3^1] = \frac{1}{5} [-1 + 2(0.156) - 3(-0.508)] = 0.167$$

$$x_2^2 = \frac{1}{9} [2 + 3x_1^2 - x_3^1] = \frac{1}{9} [2 + 3(0.167) - (-0.508)] = 0.334$$

$$x_3^2 = \frac{-1}{7} [3 - 2x_1^2 + x_2^2] = \frac{-1}{7} [3 - 2(0.167) + (0.334)] = -0.429$$

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For $m=2$

$$x_1^3 = \frac{1}{5} [-1 + 2x_2^2 - 3x_3^2] = \frac{1}{5} [-1 + 2(0.334) - 3(-0.429)] = 0.191$$

$$x_2^3 = \frac{1}{9} [2 + 3x_1^3 - x_3^2] = \frac{1}{9} [2 + 3(0.191) - (-0.429)] = 0.333$$

$$x_3^3 = \frac{-1}{7} [3 - 2x_1^3 + x_2^3] = \frac{-1}{7} [3 - 2(0.191) + (0.333)] = -0.422$$

For $m=3$

$$x_1^4 = 0.186$$

$$x_2^4 = 0.331$$

$$x_3^4 = -0.423$$

For $m=4$

$$x_1^5 = 0.186$$

$$x_2^5 = 0.331$$

$$x_3^5 = -0.423$$

Ans.Example (3)

$$x_1 - 5x_2 = -4$$

$$7x_1 - x_2 = 6$$

Solution

$$x_1 = -4 + 5x_2$$

$$x_2 = -6 + 7x_1$$

$$x_1^{m+1} = -4 + 5x_2^m$$

$$x_2^{m+1} = -6 + 7x_1^m$$

For $m=0$

$$x_1^1 = -4 + 5x_2^0 = -4 + 5(0) = -4$$

$$x_2^1 = -6 + 7x_1^0 = -6 + 7(0) = -6$$

For $m=1$

$$x_1^2 = -4 + 5x_2^1 = -4 + 5(-6) = -34$$

$$x_2^2 = -6 + 7x_1^1 = -6 + 7(-4) = -34$$

$$= -6 + 7(-4) = -34$$

For $m=2$

$$x_1^3 = -4 + 5x_2^2 = -174$$

$$x_2^3 = -6 + 7x_1^2 = -244$$

For $m=3$

$$x_1^4 = -1244$$

$$x_2^4 = -1244$$

For $m=4$

$$x_1^5 = -6124$$

$$x_2^5 = -8574$$

For $m=5$

$$x_1^6 = -42874$$

$$x_2^6 = -42874$$

Divergent

Convergence Condition

If matrix of coefficient is strictly diagonally dominant then iterative methods will always converge. That is,

$$|a_{11}| > |a_{12}| + |a_{13}| + |a_{14}| + \dots + |a_{1n}|$$

$$|a_{22}| > |a_{21}| + |a_{23}| + |a_{24}| + \dots + |a_{2n}|$$

$$|a_{33}| > |a_{31}| + |a_{32}| + \dots + |a_{3n}|$$

$$\left. \begin{array}{c} \vdots \\ |a_{nn}| > |a_{n1}| + |a_{n2}| + |a_{n3}| + \dots + |a_{nn-1}| \end{array} \right\}$$

Example

$$4x_1 + 2x_2 - x_3 = 1$$

$$x_1 + x_2 + 3x_3 = 4$$

$$3x_1 - 5x_2 + x_3 = -2$$

$$A = \begin{bmatrix} 4 & 2 & -1 \\ 1 & 1 & 3 \\ 3 & -5 & 1 \end{bmatrix}$$

$$|a_{11}| = 4, |a_{12}| = 2, |a_{13}| = 1$$

$$4 > 2 + 1$$

$$|a_{21}| = 1, |a_{22}| = 1, |a_{23}| = 3$$

$$1 \not> 1 + 3$$

(Not satisfied)

Therefore cannot guarantee on convergence of iterative schemes.

Example ② ~~Inter~~ Interchange R_2 and R_3 in previous example.

$$4x_1 + 2x_2 - x_3 = 1$$

$$3x_1 - 5x_2 + x_3 = -2$$

$$x_1 + x_2 + 3x_3 = 4$$

$$|a_{11}| > |a_{12}| + |a_{13}|, |a_{22}| > |a_{21}| + |a_{23}|, |a_{33}| > |a_{31}| + |a_{32}|$$

$$4 > 2 + 1$$

$$5 > 3 + 1$$

$$3 > 1 + 1$$

All conditions satisfied

Iterative scheme(s) will be convergent.

Eigenvalues and Eigenvectors

(9)

Review $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix}$

Characteristic Equation

$$\text{Det}[A - \lambda I] = 0$$

$$A - \lambda I = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-\lambda & 2 & 0 \\ -2 & 1-\lambda & 2 \\ 1 & 3 & 1-\lambda \end{bmatrix}$$

$$\text{Det}[A - \lambda I] = 1 - \lambda [(1 - \lambda)^2 - 6] - 2[-2(1 - \lambda) - 2] + 0 = 0$$

$$\lambda = 3, +i, -i$$

Eigenvector Equation

$$AX = \lambda X$$

$$\begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 3 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$x_1 + 2x_2 = 3x_1$$

$$-2x_1 + x_2 + 2x_3 = 3x_2 \rightarrow$$

$$x_1 + 3x_2 + x_3 = 3x_3$$

$$2x_1 - 2x_2 = 0$$

$$-2x_1 - 2x_2 + 2x_3 = 0$$

$$x_1 + 3x_2 - 2x_3 = 0$$

$$X = \begin{bmatrix} 1/2 \\ 1/2 \\ 1 \end{bmatrix}$$

Power Method

(10)

This method only find the dominant eigenvalue and the corresponding eigenvector.

Dominant Eigenvalue & Dominant Eigenvector.

Let $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$ be the eigenvalues of an $n \times n$ matrix. λ_1 is called the dominant eigenvalue if

$$|\lambda_1| > |\lambda_i| \quad i=2, 3, \dots, n$$

The eigenvector corresponding to λ_1 is called the dominant eigenvector.

Example

$$A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 0 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-\lambda & 2 & 0 \\ -2 & 1-\lambda & 2 \\ 1 & 3 & -\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (1-\lambda)[(1-\lambda)(-\lambda)-2] - 2[2\lambda-2] + 0 = 0$$

$$\lambda = 2, -1, 1$$

$\lambda = 2$ is a dominant eigenvalue.

* Not every matrix has a dominant eigenvalue.

Example $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$$\lambda_1 = 1 \quad \lambda_2 = -1$$

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\lambda_1 = 2 \quad \lambda_2 = 2 \quad \lambda_3 = 1$$

Power Method

$$X_{k+1} = AX_k$$

$$k = 0, 1, 2, \dots, n$$

Then initial approximation x_0 is non-zero and

$$x_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = [1, 1, 1, 1, \dots, 1]$$

Power Method without scaling

Power Method with scaling.

Example Use power method without scaling to approximate the dominant eigenvalue of $A = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix}$ correct to 2-D places.

$$\boxed{X_{k+1} = AX_k}$$

Solution

For $k=0$

$$X_1 = AX_0 = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -10 \\ -4 \end{bmatrix} = -4 \begin{bmatrix} -10/4 \\ 1 \end{bmatrix} = -4 \begin{bmatrix} 2.50 \\ 1.00 \end{bmatrix}$$

For $k=1$

$$X_2 = AX_1 = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} -10 \\ -4 \end{bmatrix} = \begin{bmatrix} 28 \\ 10 \end{bmatrix} = 10 \begin{bmatrix} 28/10 \\ 1 \end{bmatrix} = 10 \begin{bmatrix} 2.80 \\ 1.00 \end{bmatrix}$$

For $k=2$

$$X_3 = AX_2 = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 28 \\ 10 \end{bmatrix} = \begin{bmatrix} -64 \\ -22 \end{bmatrix} = -22 \begin{bmatrix} 2.91 \\ 1.00 \end{bmatrix}$$

For $k=3$

$$X_4 = AX_3 = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} -64 \\ -22 \end{bmatrix} = \begin{bmatrix} 136 \\ 46 \end{bmatrix} = 46 \begin{bmatrix} 2.96 \\ 1.00 \end{bmatrix}$$

For $k=4$

$$X_5 = AX_4 = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 136 \\ 46 \end{bmatrix} = \begin{bmatrix} -280 \\ -94 \end{bmatrix} = -94 \begin{bmatrix} 2.98 \\ 1.00 \end{bmatrix}$$

For $k=5$

$$X_6 = AX_5 = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} -280 \\ -94 \end{bmatrix} = \begin{bmatrix} 190 \\ 106 \end{bmatrix} = 106 \begin{bmatrix} 2.99 \\ 1.00 \end{bmatrix}$$

For $k=6$

$$X_7 = AX_6 = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 190 \\ 106 \end{bmatrix} = \begin{bmatrix} -106 \\ -280 \end{bmatrix} = -106 \begin{bmatrix} 2.99 \\ 1.00 \end{bmatrix}$$

Eigenvector $X = \begin{bmatrix} 2.99 \\ 1.00 \end{bmatrix}$

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Eigenvalue $\lambda = \frac{AX \cdot X}{X \cdot X}$ (Rayleigh Quotient)

$$\lambda = \frac{\begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 2.99 \\ 1.00 \end{bmatrix} \cdot \begin{bmatrix} 2.99 \\ 1.00 \end{bmatrix}}{\begin{bmatrix} 2.99 \\ 1.00 \end{bmatrix} \cdot \begin{bmatrix} 2.99 \\ 1.00 \end{bmatrix}}$$

$$= \frac{\begin{bmatrix} -6.02 \\ -2.01 \end{bmatrix} \cdot \begin{bmatrix} 2.99 \\ 1.00 \end{bmatrix}}{\begin{bmatrix} 2.99 \\ 1.00 \end{bmatrix} \cdot \begin{bmatrix} 2.99 \\ 1.00 \end{bmatrix}} = \frac{-20.0}{9.94} \approx \underline{\underline{-2.01}}$$

Ans.

Power Method with scaling.

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$$A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix}$$

Solution

$$X_{k+1} = AX_k$$

For $k=0$

$$X_1 = AX_0 = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix} = 5 \begin{bmatrix} 0.60 \\ 0.20 \\ 1.00 \end{bmatrix}$$

For $k=1$

$$X_2 = AX_1 = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 0.60 \\ 0.20 \\ 1.00 \end{bmatrix} = \begin{bmatrix} 1.00 \\ 1.00 \\ 2.20 \end{bmatrix} = 2.20 \begin{bmatrix} 0.45 \\ 0.45 \\ 1.00 \end{bmatrix}$$

For $k=2$

$$X_3 = AX_2 = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 0.45 \\ 0.45 \\ 1.00 \end{bmatrix} = 2.82 \begin{bmatrix} 0.48 \\ 0.55 \\ 1.00 \end{bmatrix}$$

$$X_4 = AX_3 = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 0.48 \\ 0.55 \\ 1.00 \end{bmatrix} = 3.13 \begin{bmatrix} 0.51 \\ 0.51 \\ 1.00 \end{bmatrix}$$

$$X_5 = AX_4 = \begin{bmatrix} \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } \end{bmatrix} \begin{bmatrix} 0.51 \\ 0.51 \\ 1.00 \end{bmatrix} = 3.02 \begin{bmatrix} 0.50 \\ 0.49 \\ 1.00 \end{bmatrix}$$

$$X_6 = AX_5 = \begin{bmatrix} \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } \end{bmatrix} \begin{bmatrix} 0.50 \\ 0.49 \\ 1.00 \end{bmatrix} = 2.99 \begin{bmatrix} 0.50 \\ 0.50 \\ 1.00 \end{bmatrix}$$

$$X_7 = AX_6 = \begin{bmatrix} \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } \end{bmatrix} \begin{bmatrix} 0.50 \\ 0.50 \\ 1.00 \end{bmatrix} = 3.00 \begin{bmatrix} 0.50 \\ 0.50 \\ 1.00 \end{bmatrix}$$

$$\text{Eigenvector } X = \begin{bmatrix} 0.50 \\ 0.50 \\ 1.00 \end{bmatrix}$$

$$\text{Eigenvalue } \lambda = 3.00$$